

5. JOIN Operations and Views

Perform various types of JOIN operations to extract meaningful relationships between tables. Create and manage different database views.

```
mysql> create database design;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> use design;
```

Database changed

```
mysql> CREATE TABLE Customer (customer_id INT PRIMARY KEY, customer_name  
VARCHAR(50), city VARCHAR(30));
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> CREATE TABLE Product (product_id INT PRIMARY KEY, product_name  
VARCHAR(50), price DECIMAL(10,2));
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> CREATE TABLE Orders (order_id INT PRIMARY KEY, customer_id INT, product_id  
INT, quantity INT, FOREIGN KEY (customer_id) REFERENCES  
Customer(customer_id), FOREIGN KEY (product_id) REFERENCES Product(product_id));
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> INSERT INTO Customer VALUES(1, 'Amit', 'Pune'), (2, 'Neha', 'Mumbai'), (3, 'Rahul',  
'Nagpur');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Product VALUES(101, 'Laptop', 55000), (102, 'Mobile', 20000), (103,  
'Headphones', 2500);
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Orders VALUES(1001, 1, 101, 1),(1002, 2, 102, 2),(1003, 1, 103, 3);
```

```
Query OK, 3 rows affected (0.01 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

INNER JOIN

```
mysql> SELECT customer_name, product_name, quantity FROM Orders INNER JOIN  
Customer ON Orders.customer_id = Customer.customer_id INNER JOIN Product ON  
Orders.product_id = Product.product_id;
```

```
+-----+-----+-----+  
| customer_name | product_name | quantity |  
+-----+-----+-----+  
| Amit         | Laptop       | 1        |  
| Neha         | Mobile       | 2        |  
| Amit         | Headphones   | 3        |  
+-----+-----+-----+
```

```
3 rows in set (0.00 sec)
```

LEFT JOIN

```
mysql> SELECT customer_name, order_id, product_name FROM Customer LEFT JOIN Orders  
ON Customer.customer_id = Orders.customer_id LEFT JOIN Product ON Orders.product_id =  
Product.product_id;
```

```
+-----+-----+-----+  
| customer_name | order_id | product_name |  
+-----+-----+-----+  
| Amit         | 1001    | Laptop       |  
| Amit         | 1003    | Headphones   |  
| Neha         | 1002    | Mobile       |  
| Rahul        | NULL    | NULL         |  
+-----+-----+-----+
```

4 rows in set (0.00 sec)

RIGHT JOIN

```
mysql> SELECT product_name, customer_name FROM Orders RIGHT JOIN Product ON  
Orders.product_id = Product.product_id LEFT JOIN Customer ON Orders.customer_id =  
Customer.customer_id;
```

```
+-----+-----+  
| product_name | customer_name |  
+-----+-----+  
| Laptop      | Amit          |  
| Mobile      | Neha          |  
| Headphones  | Amit          |  
+-----+-----+
```

3 rows in set (0.00 sec)

FULL OUTER JOIN

```
mysql> SELECT customer_name, product_name FROM Customer FULL OUTER JOIN Orders  
ON Customer.customer_id = Orders.customer_id FULL OUTER JOIN Product ON  
Orders.product_id = Product.product_id;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'OUTER JOIN Orders ON Customer.customer_id = Orders.customer_id FULL OUTER JOIN P' at line 1

CROSS JOIN

```
mysql> SELECT customer_name, product_name FROM Customer CROSS JOIN Product;
```

```
+-----+-----+
| customer_name | product_name |
+-----+-----+
| Rahul        | Laptop       |
| Neha         | Laptop       |
| Amit         | Laptop       |
| Rahul        | Mobile       |
| Neha         | Mobile       |
| Amit         | Mobile       |
| Rahul        | Headphones   |
| Neha         | Headphones   |
| Amit         | Headphones   |
+-----+-----+
```

9 rows in set (0.00 sec)

SELF JOIN

```
mysql> SELECT A.customer_name AS Customer1, B.customer_name AS Customer2, A.city
FROM Customer A JOIN Customer B ON A.city = B.city AND A.customer_id <>
B.customer_id;
```

Empty set (0.00 sec)

CREATE VIEW

```
mysql> CREATE VIEW Order_Details_View AS SELECT order_id, customer_name,  
product_name, quantity, price,(quantity * price) AS total_amount FROM Orders JOIN Customer  
ON Orders.customer_id = Customer.customer_id JOIN Product ON Orders.product_id =  
Product.product_id;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> show tables;
```

```
+-----+  
| Tables_in_design |  
+-----+  
| customer          |  
| order_details_view |  
| orders            |  
| product           |  
+-----+
```

4 rows in set (0.00 sec)

DISPLAY VIEW

```
mysql> SELECT * FROM Order_Details_View;
```

```
+-----+-----+-----+-----+-----+-----+  
| order_id | customer_name | product_name | quantity | price  | total_amount |  
+-----+-----+-----+-----+-----+-----+  
| 1001 | Amit          | Laptop      | 1 | 55000.00 | 55000.00 |  
| 1002 | Neha          | Mobile      | 2 | 20000.00 | 40000.00 |  
| 1003 | Amit          | Headphones  | 3 | 2500.00  | 7500.00  |  
+-----+-----+-----+-----+-----+-----+
```

3 rows in set (0.00 sec)

```
mysql> desc Order_Details_View;
```

Field	Type	Null	Key	Default	Extra
order_id	int	NO		NULL	
customer_name	varchar(50)	YES		NULL	
product_name	varchar(50)	YES		NULL	
quantity	int	YES		NULL	
price	decimal(10,2)	YES		NULL	
total_amount	decimal(20,2)	YES		NULL	

6 rows in set (0.00 sec)

```
mysql> CREATE VIEW Pune_Customers AS SELECT customer_name FROM Customer
WHERE city = 'Pune';
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> show tables;
```

Tables_in_design
customer
order_details_view
orders
product
pune_customers

5 rows in set (0.00 sec)

```
mysql> UPDATE Pune_Customers SET customer_name = 'Amit Kumar' WHERE  
customer_name = 'Amit';
```

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> select * from Pune_Customers;
```

```
+-----+  
| customer_name |  
+-----+  
| Amit Kumar   |  
+-----+
```

1 row in set (0.00 sec)

DROP VIEW

```
mysql> DROP VIEW Order_Details_View;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> show tables;
```

```
+-----+  
| Tables_in_design |  
+-----+  
| customer         |  
| orders           |  
| product          |  
| pune_customers   |  
+-----+
```

4 rows in set (0.00 sec)